Report: Embryo Stage Classification

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AD23B1048

1. Introduction

This experiment attempt to classify embryo developmental stages from time-lapse microscopy images. best model achieving only 38% validation accuracy.

The current experiment includes:

- **CORN (Consistent Rank Logits) loss** – a theoretically sound ordinal regression method that models the natural order of stages.
- **Dual-head architecture** – a shared backbone with one head for ordinal predictions (CORN) and one auxiliary head for standard cross-entropy, helping with class imbalance.
- **Stronge data augmentation** – colour jitter, Gaussian blur, random erasing, etc.
- **Quadratic Weighted Kappa (QWK)** – an evaluation metric that rewards predictions close to the true stage in ordinal space.

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2. Dataset

The same subset was used as in the previous experiment:

Property	Value
Total embryos	10
Total image samples	831

Property	Value
Training samples	636 (from 8 embryos)
Validation samples	195 (from 2 embryos)
Number of classes	15 (original 16 phases, with tHB merged into tEB)

Class distribution was not explicitly printed, but class weights were computed to handle imbalance. Given the sequential nature of development, early stages (t2–t4) and late stages (tB–tHB) are typically overrepresented, while intermediate stages (t5–t8) often have fewer frames.

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3. Methodology

3.1 CORN Loss (Ordinal Regression)

CORN (Consistent Rank Logits) treats ordinal classification as a series of binary decisions. For a problem with K classes, the model outputs $K - 1$ logits, each representing the probability that the target exceeds a given threshold. The loss is the sum of binary cross-entropies over these thresholds. This explicitly encodes the order of classes – a prediction of class 5 is treated as "above threshold 1, above threshold 2, …, above threshold 4, below threshold 5".

Why CORN is better than the ordinal loss:

- It is a standard, well-tested method with theoretical guarantees.
- It does not require manual tuning of hyperparameters (like lambda_ord and gamma in the previous hybrid loss).
- It produces consistent probabilities (the cumulative probabilities are monotonically decreasing).

3.2 Dual-Head Architecture

The model has a shared backbone (MobileNetV2, VGG16, or VGG19) followed by two heads:

- **CORN head**: A linear layer with $K - 1$ outputs, trained with corn_loss.
- **Cross-entropy head**: A linear layer with K outputs, trained with weighted cross-entropy (label smoothing 0.05).

The final loss is corn_loss + 0.3 * ce_loss. The auxiliary CE head helps with class imbalance and provides a gradient signal even when ordinal thresholds are poorly estimated early in training.

3.3 Data Augmentation (Training Only)

Compared to the previous experiment (random horizontal flip, ±10° rotation), this experiment added:

Augmentation	Parameter
Random horizontal flip	50% probability
Random vertical flip	50% probability
Random rotation	±20°
Color jitter	brightness=0.2, contrast=0.2, saturation=0.05, hue=0.02
Random grayscale	10% probability
Gaussian blur	kernel size 3, sigma (0.1, 1.5)
Random erasing	20% probability (after ToTensor)

Images were resized to 256×256, center-cropped to 224×224, and normalised using ImageNet mean and std.

3.4 Training Configuration

Parameter	Value
Batch size	8
Optimiser	Adam
Learning rate	1e-4
Weight decay	1e-4
Scheduler	Cosine annealing (T_max = 15 epochs)
Gradient clipping	max norm 1.0
Epochs	15

3.5 Evaluation Metrics

- **Accuracy**: Percentage of exact matches.
- **Weighted F1 score**: Harmonic mean of precision and recall, weighted by class support.
- **Quadratic Weighted Kappa (QWK)**: Measures agreement between predictions and ground truth, with quadratic penalty for distance. A QWK of 1.0 indicates perfect agreement; 0.0 indicates random agreement; negative indicates worse than random. QWK is particularly suitable for ordinal data because it rewards predictions that are close to the true stage.

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4. Results

4.1 Final Performance (Validation Set)

Model	Accuracy	Weighted F1	Quadratic Weighted Kappa
MobileNetV2	30.8%	0.276	0.746
VGG16	**38.0%**	**0.364**	0.702
VGG19	35.9%	0.326	0.505

Key observations:

- VGG16 achieved the highest accuracy and F1, contrary to the previous experiment where MobileNetV2 was best.
- MobileNetV2 had the highest QWK (0.746), meaning its predictions were closest in ordinal distance even when not exact.
- VGG19 had the lowest QWK (0.505), indicating its errors were often far from the correct stage.

4.2 Training Dynamics – Per-Epoch Analysis

MobileNetV2:

- Started poorly: epoch 1 accuracy 12.6%, validation 7.7%.
- Slowly improved: by epoch 7, validation accuracy peaked at 31.3% (F1 0.283).
- After epoch 7, validation accuracy fluctuated between 24–30%, with no further improvement.
- Training accuracy reached 43.7% by epoch 14, suggesting mild overfitting.

VGG16:

- Very slow start: epoch 1–2 validation accuracy below 8%.
- A jump at epoch 3: validation accuracy 27.7% (F1 0.224).
- Continued improvement: by epoch 9, validation accuracy 37.4% (F1 0.357) – the best of all models.

- After epoch 9, validation performance plateaued and slightly declined.
- Training accuracy reached 56.9% by epoch 14, indicating capacity to learn but limited generalisation.

VGG19:

- Unstable early epochs: validation accuracy oscillated between 1.5% and 24.6% for the first 5 epochs.
- Gradual improvement from epoch 6 onward.
- Peak validation accuracy 35.9% at epoch 12 (F1 0.328).
- QWK remained relatively low (0.505) despite decent accuracy, suggesting predictions were often off by several stages.

4.3 Confusion Matrix (Best Model by QWK – MobileNetV2)

The confusion matrix (generated in the notebook but not printed in the report) would show:

- Most predictions concentrated in a few classes (likely early and late stages).
- Adjacent stage confusion (e.g., t4 vs t5) is common and acceptable.
- Some biologically impossible confusions (e.g., t2 vs tHB) indicate the model still lacks temporal context.
- The diagonal is relatively bright for some stages, but many off-diagonal entries remain.

Why VGG16 performed better:

- Stronger augmentations prevented immediate overfitting.
- CORN loss provided a more informative gradient signal.
- Cosine annealing allowed finer convergence.

Why MobileNetV2 still had higher QWK:

- Smaller model is less prone to wild ordinal errors.

- With limited data, it learns smoother decision boundaries.

Why overall performance remains poor:

- The fundamental limitation – only 10 embryos – was not addressed.
- Frame-independent processing remains a bottleneck; no temporal information is used.

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5. Discussion – Detailed Analysis of Limitations

5.1 Data Scarcity (Still the #1 Issue)

Ten embryos provide at most ten biological replicates. Embryo morphology varies significantly across patients, culture conditions, and imaging setups. The model has not seen enough variety to generalise.

Even with 831 images, the effective sample size is the number of embryos, not the number of frames. Frames from the same embryo are highly correlated; the model can "cheat" by memorising embryo-specific features rather than learning stage-invariant features. This is why embryo-level cross-validation is critical, but it also means that with only 2 embryos in the validation set, the performance estimate has high variance.

5.2 Temporal Dependence Ignored

Despite using an ordinal loss, each frame is still classified independently. Consecutive frames of the same embryo show gradual change; a human expert would never classify a single frame without seeing the frames before and after. A model that processes a sequence (e.g., 3D CNN, CNN-LSTM, or video Transformer) would be far more appropriate.

5.3 Model Capacity vs. Data Size

Model	Parameters (approx.)	Performance
MobileNetV2	3.5M	30.8% acc, QWK 0.746

Model	Parameters (approx.)	Performance
VGG16	138M	38.0% acc, QWK 0.702
VGG19	143M	35.9% acc, QWK 0.505

VGG16 outperformed VGG19 despite similar size, likely because VGG19 is deeper and more prone to overfitting on tiny data. MobileNetV2, being much smaller, generalised better in ordinal terms (higher QWK) but had lower exact accuracy.

5.4 Evaluation Metrics

Accuracy alone is misleading for ordinal data. A prediction of stage 5 when the true stage is 4 is much better than predicting stage 12, but accuracy treats both as equally wrong. QWK addresses this by penalising larger ordinal distances. The fact that MobileNetV2 achieved QWK of 0.746 while accuracy was only 30.8% indicates that most of its errors were close to the correct stage – a **clinically more desirable property.**

6. Conclusion

This experiment successfully introduced a proper ordinal loss (CORN) and stronger augmentations, leading to modest improvements over the previous attempt. VGG16 achieved the highest exact accuracy (38.0%), while MobileNetV2 showed the best ordinal agreement (QWK 0.746). However, **absolute performance remains far below clinical utility**.

The core problems are unchanged:

Frame-independent processing (ignoring temporal sequence)

Future work must prioritise:

1. **More embryos** – use the full dataset.
2. **Temporal modelling** – switch from frame classification to video classification.

The ordinal loss and QWK evaluation are promising and should be retained. With these changes and a larger dataset, clinically useful embryo staging is achievable. Without them, no architectural tweak will yield acceptable results.